TOPIC : 6.0 POLYNOMIALS
SUBTOPIC : 6.1 Polynomials

Learning Outcomes:
a) Perform addition, subtraction and multiplication of polynomials
b) Perform division of polynomials

$$P(x) = ax^n + bx^{n-1} + \ldots + yx + z$$

- **All power of** x are **positive integer**
- n, the **highest power of** x represents the **degree** of polynomial.
- a, b,…,z are constant, called the non-zero coefficient

Polynomials	Degree	Name
$ax+b$, $a \neq 0$	1	Linear
ax^2+bx+c , $a \neq 0$	2	Quadratic
ax^3+bx^2+cx+d , $a \neq 0$	3	Cubic
$ax^4+bx^3+cx^2+dx+e$, $a \neq 0$	4	Quartic

Identify:

Example	Polynomial (Yes/ No)	Degree
$2x^4+3x^2-6x+11$		
$\frac{1}{4x^{-2}}+3x$		
$\frac{1}{2x^3}+3$		
$\sqrt[3]{x}+2x$		

Algebraic operations on polynomials

1. Addition and subtraction

Example 1

Given $P(x)=4x^4-10x^3-\frac{1}{3}x^2+6x-2$ and $Q(x)=-2x^4+3x^3-5x^2-2x+11$. Determine

a) $P(x)+Q(x)$ b) $Q(x)-P(x)$

Solution

a) $P(x)+Q(x)=\left(4x^4-10x^3-\frac{1}{3}x^2+6x-2\right)+\left(-2x^4+3x^3-5x^2-2x+11\right)$

$=\left(4x^4-2x^4\right)+\left(-10x^3+3x^3\right)+\left(-\frac{1}{3}x^2-5x^2\right)+\left(6x-2x\right)-2+11$ Collect the same terms

$=2x^4-7x^3-\frac{16}{3}x^2+4x+9$ Descending order degree of x

b) $Q(x)-P(x)=\left(-2x^4+3x^3-5x^2-2x+11\right)-\left(4x^4-10x^3-\frac{1}{3}x^2+6x-2\right)$

$=-2x^4+3x^3-5x^2-2x+11-4x^4+10x^3+\frac{1}{3}x^2-6x+2$ Collect the same terms

$=\left(-2x^4-4x^4\right)+\left(3x^3+10x^3\right)+\left(-5x^2+\frac{1}{3}x^2\right)+\left(-2x-6x\right)+11+2$

$=-6x^4+13x^3-\frac{14}{3}x^2-8x+13$ Descending order degree of x

2. Multiplication

Example 2

Given $P(x)=x^2-2x-1$ and $Q(x)=-2x^3+3x^2-2$. Determine

(a) $4P(x)-\frac{1}{4}Q(x)$ (b) $2P(x)Q(x)$

Solution

a) $4P(x)-\frac{1}{4}Q(x)=4\left(x^2-2x-1\right)-\frac{1}{4}\left(-2x^3+3x^2-2\right)$

$=\left(4x^2-8x-4\right)-\left(-\frac{1}{2}x^3+\frac{3}{4}x^2-\frac{1}{2}\right)$

$=4x^2-8x-4+\frac{1}{2}x^3-\frac{3}{4}x^2+\frac{1}{2}$

$=\frac{1}{2}x^3+\left(4x^2-\frac{3}{4}x^2\right)-8x-4+\frac{1}{2}$

$=\frac{1}{2}x^3+\frac{13}{4}x^2-8x-\frac{7}{2}$

Multiply 4 and $\frac{1}{4}$ into each term in the bracket

Collect the same terms

Descending order degree of x

b) $2P(x)Q(x)=2\left(x^2-2x-1\right)\left(-2x^3+3x^2-2\right)$

$=2\left(-2x^5+4x^4+2x^3+3x^4-6x^3-3x^2-2x^2+4x+2\right)$

$=2\left(-2x^5+7x^4-4x^3-5x^2+4x-2\right)$

$=-4x^5+14x^4-8x^3-10x^2+8x-4$

Expand the brackets properly

Collect the same terms

3. Division

$P(x)$: Polynomial
$Q(x)$: Quotient
$D(x)$: Divisor
$R(x)$: Remainder

Long Division Method

$$D(x)\overline{)P(x)}^{\,Q(x)} \quad \Rightarrow \quad \frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)} \qquad \textbf{OR} \qquad P(x) = D(x)Q(x) + R(x)$$

$$-$$

$$\overline{R(x)}$$

Note:
Remainder, $R(x)$ is **one-degree lower** divisor, $D(x)$.

Example 3

Divide $12 - 4x - 6x^2 + 8x^3 + 2x^4$ by $x^2 + 5$

Solution

Example 4

Determine $\dfrac{x^4+x^3+x+6}{(x+1)(x+3)}$

Solution

TOPIC : **6.0 POLYNOMIALS**
SUBTOPIC : **6.2 Remainder Theorem, Factor Theorem and Zeros of Polynomials.**

Learning Outcomes:
a) Relate the remainder and factor theorems to solve problems.
b) Find the roots of the equations and the zeros of a polynomial.

What is the remainder if polynomial $P(x)=2x^3-x^2+3x+1$ divided by $x-2$?

$$
\begin{array}{r}
2x^2+3x+9 \\
x-2\overline{)\,2x^3-x^2+3x+1} \\
\underline{-(2x^3-4x^2)} \\
3x^2+3x \\
\underline{-(3x^2-6x)} \\
9x+1 \\
\underline{-(9x-18)} \\
19
\end{array}
$$

Hence, the remainder is 19.

We can still **get the remainder without going through the long division**. This is called **the remainder theorem**. Substitute $x=2$ in the above polynomial, we obtain

$$
\begin{aligned}
P(2) &= 2(2)^3-2^2+3(2)+1 \\
&= 19
\end{aligned}
$$

Find the remainder without do the long division.

The Remainder Theorem

When a polynomial $P(x)$ is **divided** by a **linear** $(x-a)$, then the **remainder** is **constant,** $R(x)=P(a)$

Case 1: When a polynomial $P(x)$ is **divided by a linear**

1. If $P(x)$ is divided by $(x+a)$, then $R(x)=P(-a)=k$
2. If $P(x)$ is divided by $(ax+b)$, then $R(x)=P\left(-\frac{b}{a}\right)=k$, where k is a constant.

Example 1

Find the remainder when $P(x)=x^4+x^3-2x^2+4x-5$ is divided by:

(a) $x+4$ (b) $1-2x$

Solution

a) $x+4$

$D(x)=x+4=0 \Rightarrow x=-4$

Step 1: Let $D(x)=0$ to find the value of x

$P(-4)=(-4)^4+(-4)^3-2(-4)^2+4(-4)-5$

$=139$

$\therefore\ R(x)=139$

Step 2: Substitute value of x into $P(x)$ to find the remainder $R(x)$.

(b) $1-2x$

Example 2

The polynomial $ax^4 - 5x^3 + bx^2 - 7x + 1$ left a remainder of -8 when it is divided by (x-1) and a remainder of $\frac{11}{2}$ when divided by ($2x$+1), determine the values of a and b.

Solution

Case 2: When a polynomial $P(x)$ is **divided by a quadratic** expression ax^2+bx+c , then the remainder is linear function, $R(x)=ax+b$.

$$P(x)=Q(x).D(x)+R(ax+b)$$

Example 3

Determine the remainder when $P(x)=3x^4+5x^3+x+6$ is divided by x^2-1.

Example 4

When $x^4+4x^3+px^2+qx+5$ is divided by x^2-1, the remainder is $2x+3$. Find the values of p and q.

Solution

FACTOR, ROOT AND ZEROS

The Factor Theorem

If a polynomial $P(x)$ is **divided by** $(x-a)$ and the **remainder** obtained is **zero**, then $(x-a)$ is called a **factor** of the polynomial $P(x)$.

- If $(x-a)$ is a factor of P(x), then $P(a)=0$
- If $(ax+b)$ is a factor of P(x), then $P\left(-\frac{b}{a}\right)=0$

Roots and Zeros of Polynomials

- Every polynomial equation of **degree n** has exactly **n roots**. Some of these roots may be repeated.
- $x = \boldsymbol{a}$ is called a **root** of the polynomial equation $P(x) = 0$
- A **zero** of a polynomial $P(x)$ is a number $\boldsymbol{a}$ such that $P(\boldsymbol{a}) = 0$

$2x^2-9x+4=0$

Factor $(x-4)(2x-1)=0$

Roots $x=4,\ x=\frac{1}{2}$

Zeros $4,\ \frac{1}{2}$

Example 5

Determine whether the following linear functions are factors of the given polynomials:

(a) $P(x) = 3 - 7x + 5x^2 - x^3,\ D(x) = x - 3$

(b) $P(x) = 2x^3 + 3x^2 - 8x + 3,\ D(x) = 2x + 1$

Solution

a) $P(x) = 3 - 7x + 5x^2 - x^3,\ D(x) = x - 3$

Let $D(x) = x - 3 = 0 \Rightarrow x = 3$ ← **Step 1**: Let $D(x) = 0$ to find the value of x

$$P(3) = 3 - 7(3) + 5(3)^2 - (3)^3$$
$$= 0$$

Step 2: Substitute value of x into $P(x)$ to find the remainder $R(x)$.

$\therefore$ Since $P(3) = 0$, thus $(x - 3)$ is a factor of $P(x)$. ← **Step 3**: Conclusion

b)

Example 6

The polynomial $2x^3 + ax^2 + bx + 8$ has a factor $(x-1)$ and gives a remainder of 50 when divided by $(x-3)$. Determine *a* and *b*.

Solution

Example 7

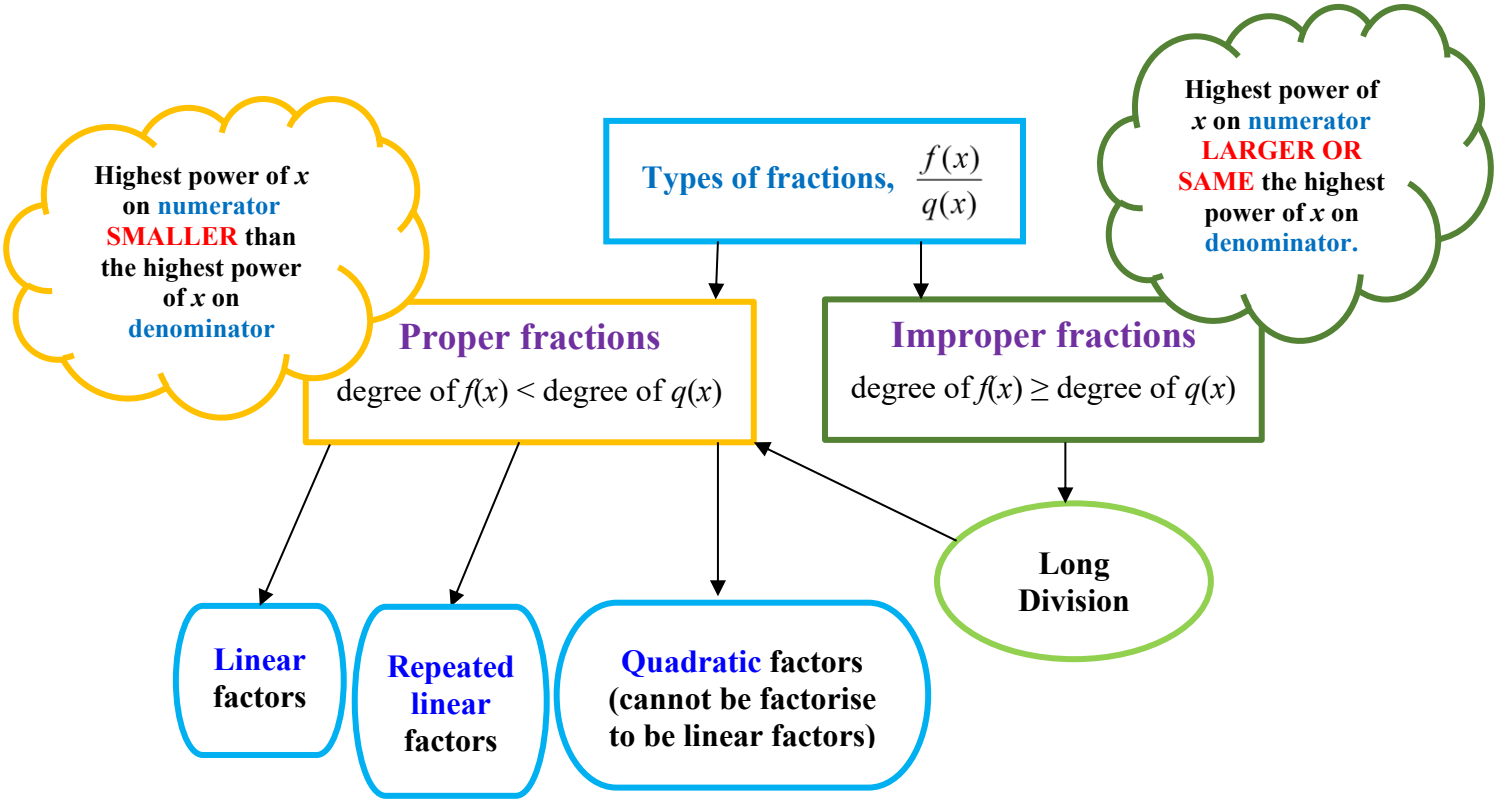
Factorize $P(x) = 2x^3 - 9x^2 + 3x + 4$ and write all the zeros.

Solution

TOPIC : 6.0 POLYNOMIALS
SUBTOPIC : 6.3 Partial fractions

Learning Outcomes:

a) Construct partial fractions decomposition when the denominators in the form of
   i. linear factor
   ii. Repeated linear factor
   iii. Quadratic factor
b) Change the rational polynomials to proper fraction when the degree of the numerator is same or more than the denominator
c) Determine the partial fractions



Identify the types of fractions:

$\frac{3x+4}{(x-1)(2x-3)}$:

$\frac{2x}{x^2-1}$:

$\frac{3x^3+4}{4-x^2}$:

$\frac{x^2+2}{x(x-1)}$:

A. Proper fractions

3 types of factors (Refer on the denominator):

Highest power of *x* on numerator SMALLER than the highest power of *x* on denominator.

1) Linear factor $\dfrac{P(x)}{(ax+b)(cx+d)}$

2) Repeated linear factors $\dfrac{P(x)}{(ax+b)^2}$

3) Quadratic factor (cannot be factorized) $\dfrac{P(x)}{\left(ax^2+bx+c\right)}$

Example of factors	Types of factors
$(2x+3)(1-x)$	
x^2+3x	
$(x-2)\left(x^2-3x+2\right)$	
x^2-1	
$x^2+3x+12$	
$x\left(x^2+2x\right)$	

Case 1: Denominator with distinct linear factors

$$\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{(ax+b)} + \frac{B}{(cx+d)}$$

Example 1

Write $\dfrac{2x+3}{(x-1)(x+2)}$ in its partial fractions form.

Solution

Example 2

Express $\dfrac{5x+1}{(x^2+x)(2x-1)}$ in partial fractions.

Solution

Case 2: Denominator with repeated linear factors

$$\frac{f(x)}{(ax+b)^2} = \frac{A}{(ax+b)} + \frac{B}{(ax+b)^2}$$

Example 3

Express $\frac{7x}{(x+2)^2}$ in partial fraction.

Solution

Example 4

Express $\frac{5x^2+3x+8}{(1-x^2)(1+x)}$ in partial fraction.

Solution

Case 3: Denominator with quadratic factors

$$\frac{f(x)}{g(x)} = \frac{Ax+B}{\left(ax^2+bx+c\right)}$$

Note: quadratic factor that **cannot be factorized to be 2 linear factors**

$$ax^2+bx+c \neq (ax+h)(x+k)$$

Example 5

Express $\dfrac{4x+1}{(x+1)\left(x^2+5x+1\right)}$ in a partial fraction.

Solution

Example 6

Write $\dfrac{3}{(x+2)\left(x^2+1\right)}$ in partial fraction.

Solution

B. Improper fraction

When the degree of $f(x)$ is **greater or equal** to the degree of $g(x)$, it is an improper fraction. Therefore, you must do the **long division** before the partial fraction.

Example 7

Express $\dfrac{x^3 - x - 5}{(x+2)(x^2+1)}$ as a partial fraction.

Solution

Example 8

Express $\frac{x^3}{x(x+1)^2}$ as a partial fraction.

Solution